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AUG 29 2016

10 CFR 50.73

Serial: BSEP 16-0077

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555

Subject: Brunswick Steam Electric Plant, Unit No. 2
Renewed Facility Operating License No. DPR-62
Docket No. 50-324
Licensee Event Report 2-2016-002

In accordance with the Code of Federal Regulations, Title 10, Part 50.73, Duke Energy Progress, Inc., submits the enclosed Licensee Event Report (LER). This report fulfills the requirement for a written report within sixty (60) days of a reportable occurrence.

This document contains no regulatory commitments.

Please refer any questions regarding this submittal to Mr. Lee Grzeck, Manager – Regulatory Affairs, at (910) 457-2487.

Sincerely,

A handwritten signature in black ink, appearing to read "WRG", written over a horizontal line.

William R. Gideon

MAT/mat

Enclosure: Licensee Event Report

cc (with enclosure):

U. S. Nuclear Regulatory Commission, Region II
ATTN: Ms. Catherine Haney, Regional Administrator
245 Peachtree Center Ave, NE, Suite 1200
Atlanta, GA 30303-1257

U. S. Nuclear Regulatory Commission
ATTN: Ms. Michelle P. Catts, NRC Senior Resident Inspector
8470 River Road
Southport, NC 28461-8869

U. S. Nuclear Regulatory Commission
ATTN: Mr. Andrew Hon (Mail Stop OWFN 8G9A) **(Electronic Copy Only)**
11555 Rockville Pike
Rockville, MD 20852-2738

Chair - North Carolina Utilities Commission **(Electronic Copy Only)**
4325 Mail Service Center
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LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Privacy and Information Collections Branch (T-5 F53), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to InfoCollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME

Brunswick Steam Electric Plant (BSEP), Unit 2

2. DOCKET NUMBER

05000324

3. PAGE

1 OF 4

4. TITLE

High Pressure Coolant Injection System Inoperable due to Failed Relay Coil

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED			
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER		
07	05	2016	2016	- 002	- 00	08	29	2016	FACILITY NAME	DOCKET NUMBER		
										05000		
9. OPERATING MODE			11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)									
1			☐ 20.2201(b)			☐ 20.2203(a)(3)(i)			☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)	
			☐ 20.2201(d)			☐ 20.2203(a)(3)(ii)			☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)	
			☐ 20.2203(a)(1)			☐ 20.2203(a)(4)			☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)	
			☐ 20.2203(a)(2)(i)			☐ 50.36(c)(1)(i)(A)			☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)	
10. POWER LEVEL			☐ 20.2203(a)(2)(ii)			☐ 50.36(c)(1)(ii)(A)			☐ 50.73(a)(2)(v)(A)		☐ 73.71(a)(4)	
			☐ 20.2203(a)(2)(iii)			☐ 50.36(c)(2)			☐ 50.73(a)(2)(v)(B)		☐ 73.71(a)(5)	
			☐ 20.2203(a)(2)(iv)			☐ 50.46(a)(3)(ii)			☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(1)	
			☐ 20.2203(a)(2)(v)			☐ 50.73(a)(2)(i)(A)			☒ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(i)	
			☐ 20.2203(a)(2)(vi)			☐ 50.73(a)(2)(i)(B)			☐ 50.73(a)(2)(vii)		☐ 73.77(a)(2)(ii)	
			☐ 50.73(a)(2)(i)(C)			☐ OTHER		Specify in Abstract below or in NRC Form 366A				

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT

Lee Grzeck, Manager – Regulatory Affairs

TELEPHONE NUMBER (Include Area Code)

(910) 457-2487

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX
X	BJ	CL	G080	Y					

14. SUPPLEMENTAL REPORT EXPECTED

☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☒ NO

15. EXPECTED SUBMISSION DATE

MONTH	DAY	YEAR

ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)

On July 5, 2016, at 1640 Eastern Daylight Time (EDT), the Unit 2 High Pressure Coolant Injection (HPCI) system was declared inoperable due to apparent failure of the HPCI Auxiliary Oil Pump. The HPCI Auxiliary Oil Pump provides hydraulic pressure required to open the HPCI Turbine Stop Valve and the HPCI Turbine Control Valve during initial HPCI startup. Failure of the HPCI Auxiliary Oil Pump prevents the HPCI system from performing its safety function. This condition is being reported in accordance with 10 CFR 50.73(a)(2)(v)(D), as an event or condition that could have prevented the fulfillment of the safety function of a system that is needed to mitigate the consequences of an accident.

The HPCI system inoperability was due to loss of control power for the Auxiliary Oil Pump. The loss of control power was caused by a failed motor overload alarm relay coil; which caused current flow in excess of the control power fuse rating. The HPCI system was restored on July 6, 2016, at 1050 EDT. The alarm relay coil failure was age related. The corrective actions include creation of preventive maintenance tasks to replace the Unit 1 and 2 HPCI motor overload alarm relay coils on an appropriate frequency.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Brunswick Steam Electric Plant (BSEP), Unit 2	05000-324	2016	- 002	- 00

NARRATIVE

Energy Industry Identification System (EIIIS) codes are identified in the text as [XX].

Background*Initial Conditions*

At the time of this event, Unit 2 was in Mode 1 at approximately 100 percent of rated thermal power. The Reactor Core Isolation Cooling (RCIC) system [BN], Automatic Depressurization System (ADS), Low Pressure Coolant Injection (LPCI) system [BO], and Core Spray (CS) system [BM] were operable at the time of the event.

Reportability Criteria

This condition is being reported in accordance with 10 CFR 50.73(a)(2)(v)(D), as an event or condition that could have prevented the fulfillment of the safety function of a system that is needed to mitigate the consequences of an accident. The NRC was initially notified of this event on July 5, 2016 (i.e., Event Number 52064).

Event Description

On July 5, 2016, at 1640 Eastern Daylight Time (EDT), the Unit 2 High Pressure Coolant Injection (HPCI) system [BJ] was declared inoperable after the control room received annunciator 2-A-1 (3-3), "HPCI Aux Oil Pump Motor Overload." Upon further investigation, it was determined that the HPCI Auxiliary Oil Pump motor overload alarm relay (i.e., 2-2XDA-B11-74) had failed and blown the associated control circuit fuse (i.e., 2-2XDA-B11-2FU). The HPCI Auxiliary Oil Pump provides hydraulic pressure required to open the HPCI Turbine Stop Valve and the HPCI Turbine Control Valve during initial HPCI startup. With failure of the relay, the HPCI Auxiliary Oil Pump was incapable of starting; rendering the HPCI system inoperable.

Operability of the HPCI system was restored on July 6, 2016, at 1050 EDT.

Event Cause

The cause of the Unit 2 HPCI inoperability was determined to be age related failure of HPCI Auxiliary Oil Pump motor overload alarm relay (i.e., 2-2XDA-B11-74) coil.

The failed relay is normally energized and was original plant equipment. The equipment reliability classification was "Important" as opposed to "Critical." Based on this classification, preventive maintenance had not been performed to replace the Unit 2 relay.

The Unit 1 HPCI Auxiliary Oil Pump motor overload alarm relay was last replaced in January 1995.

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Brunswick Steam Electric Plant (BSEP), Unit 2	05000-324	2016	- 002	- 00

NARRATIVESafety Assessment

The safety significance of this event was minimal. The RCIC system, ADS, LPCI system, and CS system were all operable at the time of the event. The HPCI system was restored to operable status within the 14-day Technical Specification Completion Time.

Corrective Actions

Any changes to the corrective actions and schedules noted below will be made in accordance with the site's corrective action program.

The following corrective actions have been completed or are planned.

- Operability of the HPCI system was restored on July 6, 2016, at 1050 EDT.
- Preventive Maintenance tasks to replace the Unit 1 and Unit 2 HPCI Auxiliary Oil Pump motor overload relays, on an appropriate frequency, will be created. This is currently scheduled to be completed by November 4, 2016.
- The equipment reliability classifications for the Unit 1 and Unit 2 HPCI Auxiliary Oil Pump motor overload relays will be changed to "Critical." This is currently scheduled to be completed by November 11, 2016.
- A review of control wiring diagrams for Mitigating Systems Performance Index (MSPI) monitored components will be performed to identify components with "Important" equipment reliability classification. This review will either document classification is correct or change to critical if appropriate and implement appropriate preventive maintenance strategies. This is currently scheduled to be completed by December 15, 2016.

Previous Similar Events

A review of LERs and condition reports for the past five years identified the following previous similar occurrences.

- LER 1-2015-001, Revision 1, dated December 21, 2015, reported inoperability of the Unit 1 HPCI system due to a faulty magnetic motor contactor coil (i.e., M-coil) within the HPCI turbine auxiliary oil pump motor contactor 1-1XDA-B11-M. The failure of the coil was determined to be most likely long term thermal cycling. The coil was replaced and the HPCI system was returned to operable status on February 20, 2015. Based on the age of the M-coil, a one-time replacement is expected to prevent recurrence of this failure through the end of licensed operation for BSEP. Therefore, no periodic replacement was recommended. Corrective actions associated with this LER were component level/function level and, therefore, would not be expected to prevent recurrence of HPCI system inoperability.



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		2016	- 002	- 00

NARRATIVE

- LER 1-2012-004, dated June 29, 2012, reported HPCI system inoperability due to the erratic HPCI governor operation caused by a failed ramp generator signal converter (RGSC) operational amplifier. The root cause of the RGSC failure is the lack of a replacement preventive maintenance (PM) task for the RGSC. The corrective action to prevent recurrence for this event was to establish the appropriate PM for periodic replacement of the RGSC on both units' HPCI systems. Corrective actions associated with this LER were component level/function level and, therefore, would not be expected to prevent recurrence of HPCI system inoperability.

Commitments

No regulatory commitments are contained in this report.